

OPERATING SYSTEMS (CS304)

5 Marks Question Bank

UNIT 1 – Processes, Threads & System Calls

1. Define Process and explain its components.
2. Explain Process States with a diagram.
3. What is a Process Control Block (PCB)?
4. Define Thread and explain its advantages.
5. Differentiate User-Level Threads and Kernel-Level Threads.
6. What are System Calls? Explain their types.

UNIT 2 – CPU Scheduling

1. Define CPU Scheduling.
2. Explain FCFS Scheduling Algorithm.
3. Explain Shortest Job First (SJF) Scheduling.
4. What is Round Robin Scheduling?
5. Define Context Switching.
6. Discuss Scheduling Criteria used in CPU Scheduling.

UNIT 3 – Deadlocks

1. Define Deadlock.
2. Explain the necessary conditions for Deadlock.
3. What is a Resource Allocation Graph?
4. Explain Deadlock Prevention.
5. What is Deadlock Avoidance?
6. Write a note on Banker's Algorithm.

UNIT 4 – Memory Management

1. Define Paging.
2. Explain Segmentation.
3. What is Virtual Memory?
4. Define Thrashing.
5. Explain Page Tables.
6. Differentiate Paging and Segmentation.

UNIT 5 – File Systems & Disk Scheduling

1. Explain File Allocation Methods.
2. What is a Directory Structure?
3. Explain FCFS Disk Scheduling.
4. What is SSTF Scheduling?
5. Explain SCAN Disk Scheduling.
6. Differentiate SCAN and C-SCAN.

Exam Preparation Tips

Focus on Process Management, CPU Scheduling Algorithms, Deadlocks, Paging, Segmentation, File Systems, and Disk Scheduling Algorithms as these topics are frequently asked in university examinations.